

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Case Study 4: Investigating Rhion Possession Evidence

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Course Title: Computer Forensics Case Studies

Course Code: SBT-DF204

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This report presents an offline examination of the supplied training evidence for Case Study 4: a 247 MiB USB disk image (RHINOUSB.dd) and three network captures (rhino.log, rhino2.log and rhino3.log). The scenario is fictional. All four originals were hashed and set read-only before analysis, and each finding below rests on a working copy and a cited record (file, hash, packet, stream or screenshot).

Principal findings

1. **Disk image.** The image is a bare FAT16 volume with no partition table. It holds two allocated files, GUMBO1.TXT and GUMBO2.TXT, which contain recipes, and no deleted directory entries were listed. File carving with PhotoRec recovered further content, including six unique JPEG images (one carved twice), two GIF images and a Word document whose metadata does not connect it to the case.
2. **FTP (rhino.log).** On 2004-04-26 between 22:21 and 22:26 UTC the account `gnome` on 137.30.122.253 uploaded three objects to the FTP server 137.30.120.40: rhino1.jpg (65,703 bytes), rhino3.jpg (96,899 bytes, sent twice, the first time in ASCII mode with a server warning) and contraband.zip (230,566 bytes). The reassembled archive is complete and its single entry, rhino2.jpg, is encrypted
3. **HTTP (rhino2.log and rhino3.log).** On 2004-04-28 the endpoint 137.30.123.234 downloaded rhino4.jpg and rhino5.gif from www.cs.uno.edu (about 21:08 UTC) and, after Google searches for "rhino.exe", a Windows executable named rhino.exe (145,920 bytes, about 21:16 UTC). The executable was hashed only. It was not run.
4. **Hidden content.** The steganography detector could not be built on the Kali VM, so no detection or recovery result is reported. Metadata, binwalk and strings checks found nothing embedded in the files tested, which is inconclusive for hiding methods that alter the image data itself.
5. **Correlation.** No FTP or HTTP image matches a carved image by SHA-256. The artefacts do not show that files on the USB device were the files transferred, and they do not identify who operated any device.

Confidence. High confidence in what the captures record (sessions, packets, headers and object bytes, each checkable by packet number and hash). Low confidence in any link between the USB image and the network activity. This report makes no person-level attribution because the evidence does not support one.

EVIDENCE ACQUISITION AND INTEGRITY

Evidence inventory

Table 1 lists each supplied original and its working copy. SHA-256 values were recorded with `sha256sum` before any other operation and again after copying (Figure 1, Figure 2 and Figure 3).

ID	File	Bytes	SHA-256	Role and location
EVID-01	RHINOUSB.dd	259,506,176	`ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65`	Original USB disk image, mode 444, ~/Downloads/LAB4SBT204/
EVID-02	RHINOUSB_My_COPY.dd	259,506,176	`ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65`	Working copy made with dd; identical hash. ~/Downloads/LAB4SBT204/working/

EVID-03	rhino.log	3,187,907	64e6d55b76660eb3aaa41572c1d04e4452510a343bbf42e844424827ddfddb2	Original FTP capture, mode 444
EVID-04	rhino2.log	292,604	41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde	Original HTTP capture, mode 444
EVID-05	rhino3.log	226,094	7b0304f5e88a30c305a99b5a1e2977bced5b9c94da6f458f887bb39966bbfc46	Original HTTP capture, mode 444
EVID-06 to 08	rhino.log, rhino2.log, rhino3.log	as above	identical to EVID-03 to 05	Working copies made with cp -p. ~/Downloads/LAB4SBT204/wo

Table 1. Evidence inventory. Source of all items: E-Campus course package, supplied for this case study.

Preservation steps and precautions

- Identification: `file` reports RHINOUSB.dd as a FAT (16 bit) boot-sector image (OEM-ID “mkdosfs”, 8 sectors per cluster, 512 root entries) and the three logs as pcap capture files, version 2.4, microsecond timestamps (Figure 1).
- Read-only originals: `chmod 444` was applied to all four originals and confirmed with `ls -l` (Figure 1).
- Working copies: the disk image was copied with `dd` (506,848 records, 259,506,176 bytes) and the captures with `cp -p`. The SHA-256 of each copy equals the original (Figure 2, Figure 3).
- Read-only Sleuth Kit commands (`fdisk`, `mmls`, `fsstat`, `fls`, `istat`, `icat`) were run against the original image before the working copy existed. They only read the file, and the SHA-256 of the original recomputed afterwards was unchanged (Figure 2). Carving, metadata review, Wireshark and tshark work used working copies.
- Tools and versions seen in the output: PhotoRec 7.2 (February 2024) (Figure 11), ExifTool
- Network and execution safeguards: no connection was made to any host named in the evidence; rhino.exe was never executed; no evidence file was uploaded anywhere; FTP credentials are redacted in screenshots and were not reused. The VM had internet access only to fetch analysis tools (stegdetect source from GitHub and the rockyou.txt wordlist).

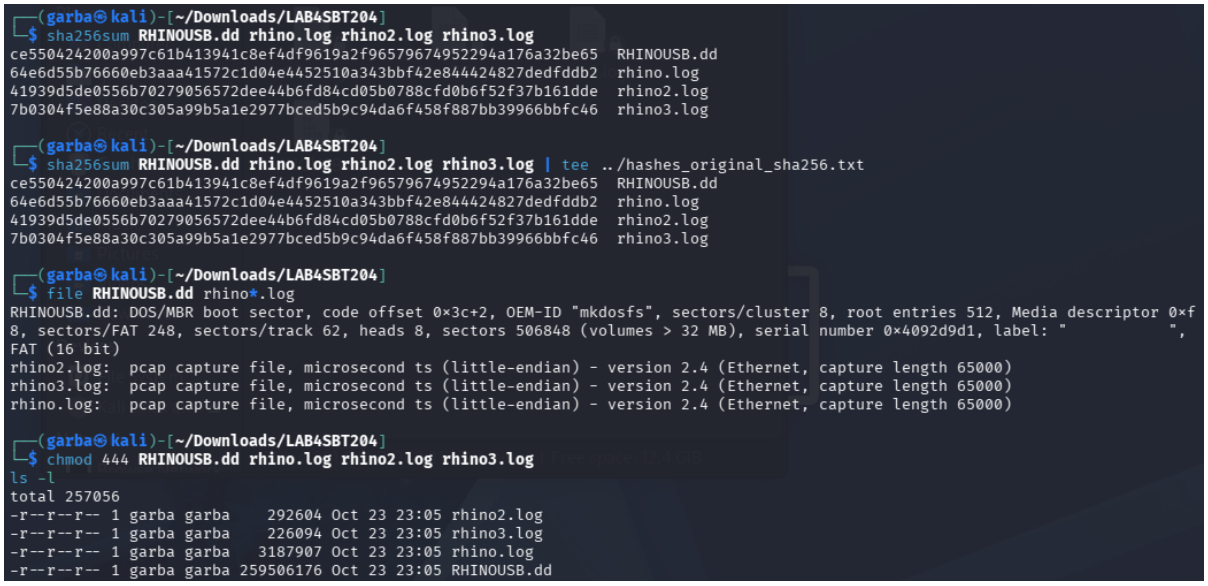


Figure 1: SHA-256 values of the four original files, file-type check, chmod 444 and read-only listing (E01).

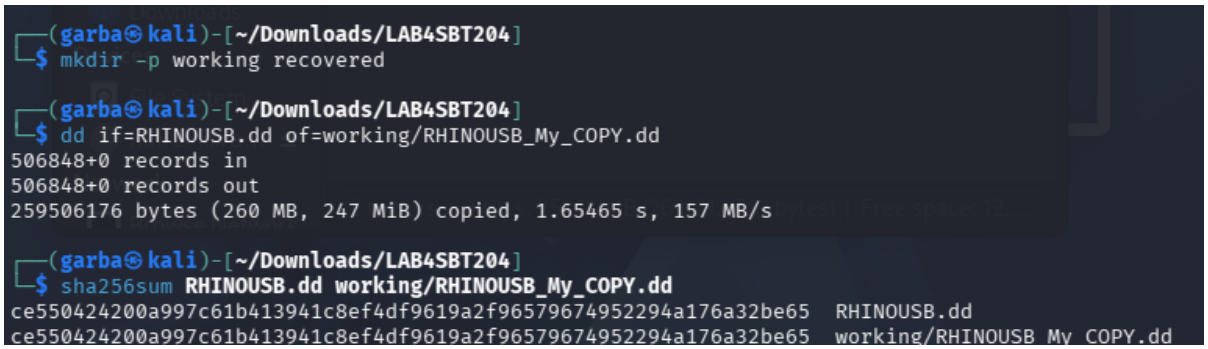


Figure 2: Working copy of the disk image created with dd; SHA-256 of original and copy identical (E02).

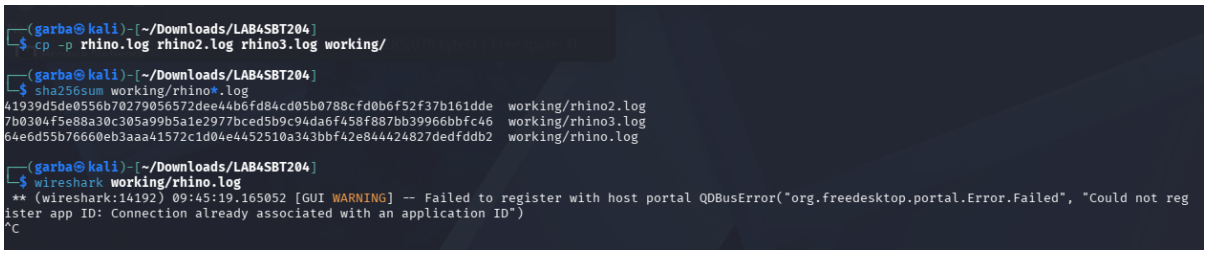


Figure 3: Working copies of the three captures (cp -p) with SHA-256 values matching the originals (E02).

Handling error and verification

While saving an extracted FTP object, the command `mv ~/Downloads/LAB4SBT204/rhino3.log ~/Downloads/LAB4SBT204/exports/ftp_contraband.zip` moved and renamed the original rhino3.log into the exports folder (Figure 4). Two further `mv` commands naming rhino2.log and rhino.log stopped at the overwrite prompt, and both files remained in place. The file was moved back and the SHA-256 of all three originals was recomputed: every value equals Table 1, and the sizes and read-only modes are unchanged (Figure 5). A move or rename does not alter file content, and the working copy of rhino3.log was unaffected throughout.

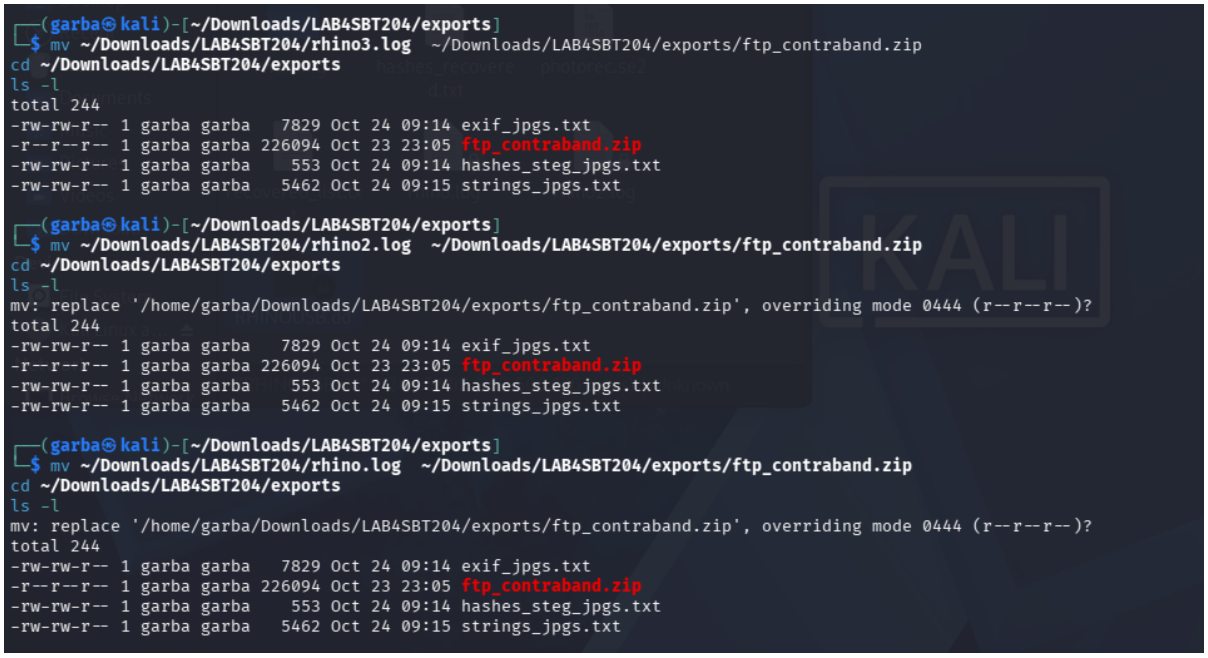


Figure 4: Handling error: original rhino3.log moved and renamed into exports/ (two later mv commands stopped at the overwrite prompt) (E21).

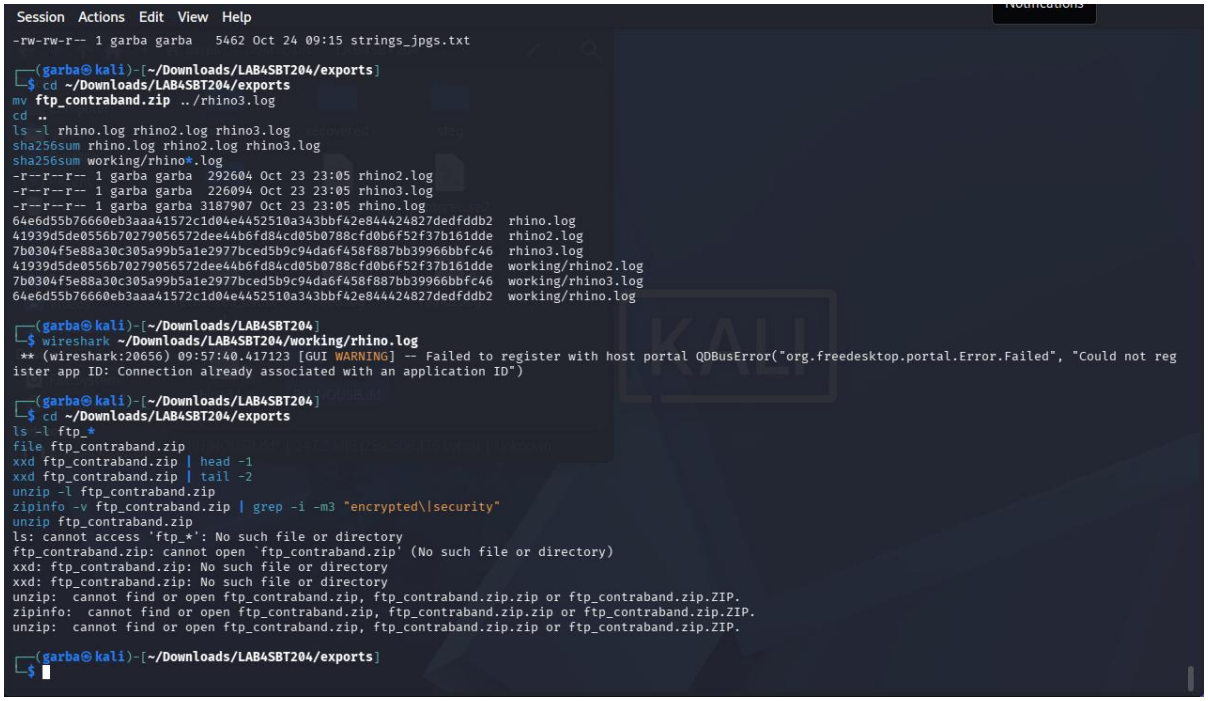


Figure 5: Original rhino3.log moved back; SHA-256 of all three originals and working copies re-verified (E21).

METHOD

The examination followed the order in the assessment brief: preserve the evidence, examine the disk image, recover files, test for hidden content, analyse the FTP and HTTP captures, validate every reconstructed object, normalise times to UTC and correlate the artefacts.

Time handling

Packet times were read with Wireshark's "UTC Date and Time of Day" format and tshark's `frame.time_utc` field. FAT file-system times carry no time zone, so they are quoted exactly as Sleuth Kit displays them and are not converted. Times written by the analysis VM (`ls -l` and PhotoRec output) are local time at an offset of +01:00, as ExifTool reports for the recovered files (Figure 15), and are converted to UTC in the timeline.

Disk-image review

Sleuth Kit tools were used: `fdisk -l` and `mmls` for partitioning, `fsstat` for file-system layout, `fls -r` for the directory tree (deleted entries would be marked with an asterisk), `istat` for per-file metadata and `icat` to read file content.

File recovery

PhotoRec 7.2 was run on the working copy. Each output file was hashed with `sha256sum`, typed with `file`, inspected with `exiftool` and compared by hash to find duplicates (`sha256sum recup_dir.*/* | sort | uniq -w64 -D`).

Hidden-content testing

Metadata (`exiftool`), appended or embedded signatures (`binwalk`) and readable strings (`strings -n 8`) were checked on the carved JPEGs. A statistical steganography detector (stegdetect, built from source on Kali) was attempted. The build failed, so that step is reported as a limitation and not as a result.

FTP analysis (rhino.log)

Display filters `ftp` and `ftp.request.command == "STOR"` located each transfer. In active-mode FTP the `PORT h1,h2,h3,h4,p1,p2` command names the data port as p1 × 256 + p2; for example `PORT 137,30,122,253,6,121` gives port 1657. The data connection was found by that port, followed with Follow TCP Stream (one direction, "Raw") and saved. Each object was compared with the byte count in the server's own transfer report.

HTTP analysis (rhino2.log, rhino3.log)

The filters `http.request.method == "GET"` and `http.response` listed the requests and responses. tshark (`-T fields` with `frame.time_utc`, and `--export-objects http`) produced UTC request lists and exported the response bodies. The exported size was compared with the response's Content-Length.

Object validation

Every reconstructed object was checked by (1) size against the server's byte count or the Content-Length, (2) format signature and trailer (JPEG `FFD8`…`FFD9`, GIF `GIF8`…`3B`, ZIP `PK` with an end-of-central-directory record, Windows executable `MZ`), (3) `file` and (4) SHA-256.

FINDINGS

Summary of answers

Q	*Finding*	*Evidence*
Q1	Four originals hashed and set read-only; working copies hash-identical; one handling error corrected and re-verified.	E01, E02, E21
Q2	Bare FAT16 volume with no partition table; two allocated files (GUMBO1.TXT, GUMBO2.TXT, recipes); no deleted directory entries listed.	E03 to E05
Q3	PhotoRec recovered 132 files in the first run: six unique JPEGs, two GIFs, one Word document, one duplicate JPEG pair.	E06 to E08
Q4	No embedded data or anomalous metadata in the files tested; stegdetect build failed, so detection was not performed.	E09 to E11
Q5	Three FTP sessions on 2004-04-26 UTC: uploads of rhino1.jpg, rhino3.jpg (twice) and contraband.zip by account gnome from 137.30.122.253.	E12 to E14
Q6	contraband.zip is complete (230,566 bytes) with one encrypted entry, rhino2.jpg.	E15, E16
Q7	rhino4.jpg, rhino5.gif and rhino.exe downloaded on 2004-04-28 from www.cs.uno.edu; rhino.exe not executed.	E17 to E20
Q8	Transfers are established; no hash link between the USB image and the network objects; no attribution to a person.	All

Q1. How did you preserve the case materials?

Finding. Four originals were identified by name, size and SHA-256 before any analysis, set read-only (mode 444) and copied. Every copy has the same SHA-256 as its original (Table 1, Figure 1 to Figure 3). Table 1 is the baseline for every later integrity check, and Figure 5 shows a re-verification after one handling error (section 3.3).

Evidence. E01, E02, E21.

Limit. A matching hash shows that a copy equals the file supplied. It does not show that the package was complete or unaltered before it reached the examiner.

Q2. What does the disk image show before any recovery?

Finding. The image holds one FAT16 volume that starts at sector 0, with no partition table. Two allocated files are visible and both contain recipes. No file-system record points to any image file.

Field	*Value*	*Source*
Partition table	None. `fdisk -l` reports a DOS disklabel (identifier 0x00000000) and lists no partitions; `mmls` printed nothing.	Figure 6
File system	FAT16, OEM name “mkdosfs”, volume ID 0x4092d9d1, no volume label, 0 sectors before the file system.	Figure 6
Geometry	512-byte sectors, 8 sectors per cluster (4,096-byte clusters), 506,848 sectors in total.	Figure 1, Figure 6
Layout (sectors)	Reserved 0 to 0; FAT 0 1 to 248; FAT 1 249 to 496; data area 497 to 506,847; root directory 497 to 528.	Figure 6
Directory listing	`fls -r`: inode 4 gumbo1.txt and inode 6 gumbo2.txt, plus Sleuth Kit virtual entries $MBR, $FAT1, $FAT2 and $OrphanFiles. No deleted entries were listed.	Figure 7

GUMBO1.TXT	2,815 bytes; Written and Created 2004-04-30 18:11:20; Accessed 2004-04-30 00:00:00; sectors 529 to 534.	Figure 7
GUMBO2.TXT	1,293 bytes; Written and Created 2004-04-30 18:11:24; Accessed 2004-04-30 00:00:00; sectors 537 to 539.	Figure 8
Content	GUMBO1.TXT is a recipe for shrimp and tasso gumbo; GUMBO2.TXT is a recipe for shrimp and andouille sausage gumbo. Neither is case-relevant.	Figure 9, Figure 10

Table 2. File-system findings for RHINOUSB.dd.

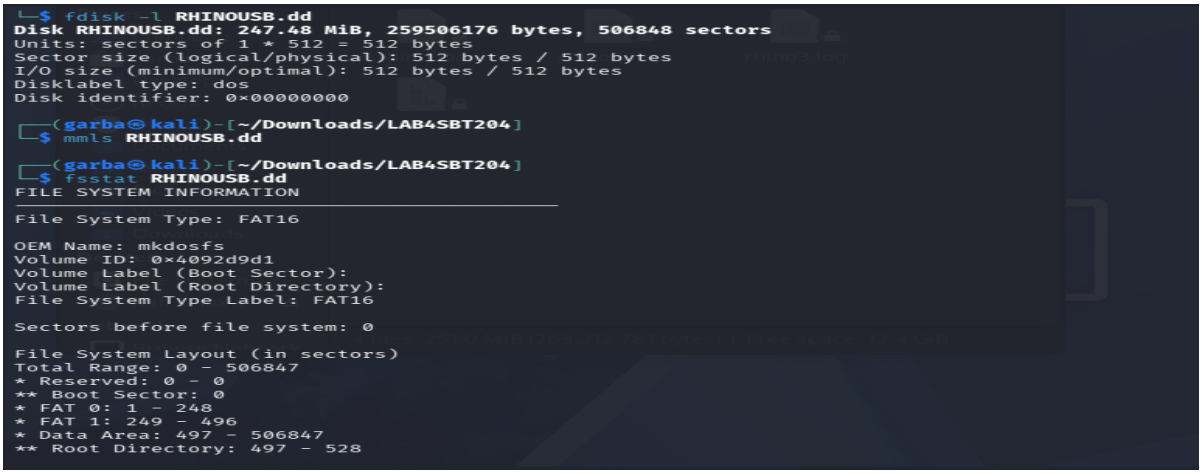


Figure 6: fdisk -l, mmls and fsstat on RHINOUSB.dd (E03).

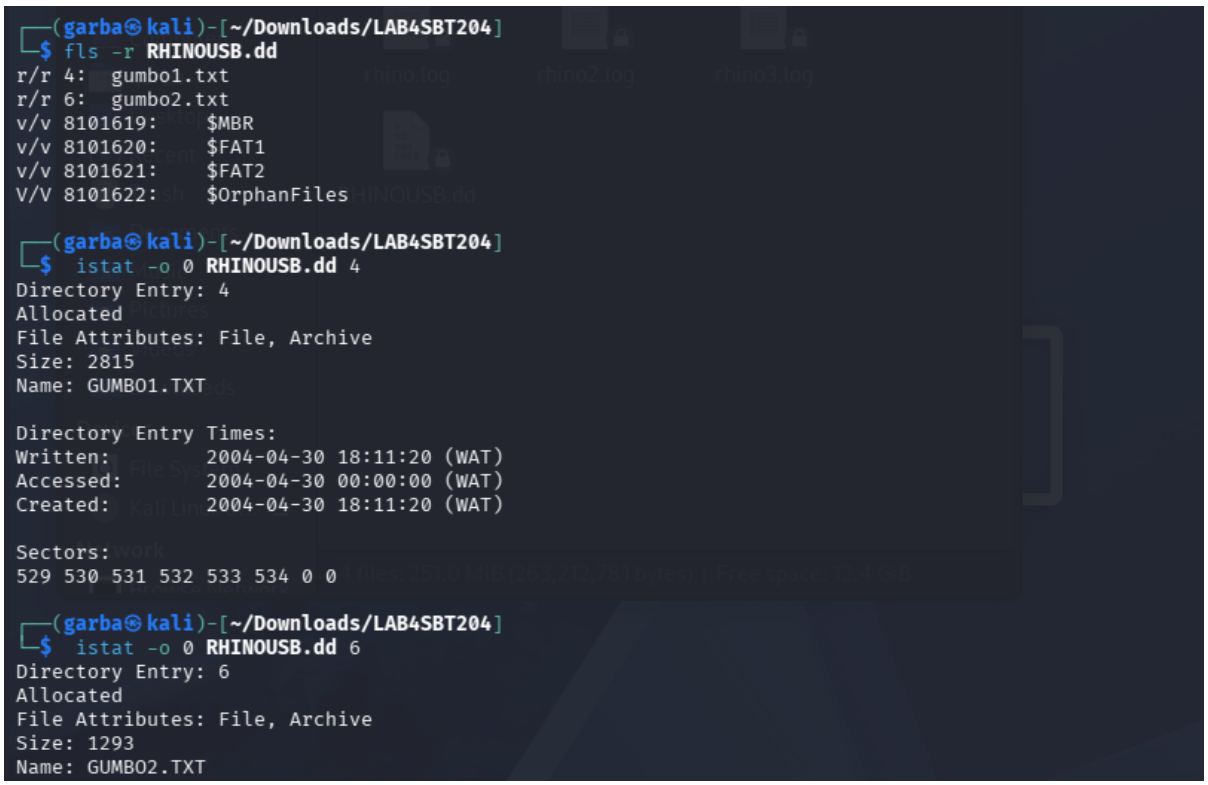


Figure 7: fls -r listing and istat for inode 4 (GUMBO1.TXT) (E04).

```
Sectors:
529 530 531 532 533 534 0 0

┌──(garba㉿kali)-[~/Downloads/LAB4SBT204]
└─$ istat -o 0 RHINOUSB.dd 6
Directory Entry: 6
Allocated
File Attributes: File, Archive
Size: 1293
Name: GUMBO2.TXT

Directory Entry Times:
Written:        2004-04-30 18:11:24 (WAT)
Accessed:       2004-04-30 00:00:00 (WAT)
Created:        2004-04-30 18:11:24 (WAT)

Sectors:
537 538 539 0 0 0 0 0

┌──(garba㉿kali)-[~/Downloads/LAB4SBT204]
└─$ icat RHINOUSB.dd 4 | less

zsh: done       icat RHINOUSB.dd 4 |
zsh: suspended  less

┌──(garba㉿kali)-[~/Downloads/LAB4SBT204]
└─$ icat RHINOUSB.dd 6 | less

zsh: done       icat RHINOUSB.dd 6 |
zsh: suspended  less
```

Figure 8: istat for inode 6 (GUMBO2.TXT) and icat of both files (E04).

```
SHRIMP AND TASSO GUMBO

Associate Food Editor: Alexis Touchet
Father: Rodney Miller, Abbeville, LA

When I was a kid, our summer routine started with the opening day of
shrimp season <97> my dad would get the boat ready and out we would go,
winding our way down the bayou until we reached Vermilion Bay. He
would check to make sure the trawl net was securely tied before
tossing it into the water, and would then let it be dragged down
slowly along with the trawl boards. When we painstakingly pulled the
net up onto the boat and untied the bag, it spilled out a catch of
shrimp, crabs, and fish. These fresh shrimp were the stars in Dad's
favorite gumbo recipe.

2 lb large shrimp in shell (21 to 25 per lb), peeled and shells reserved
14 cups water
1/4 cup vegetable oil
1/2 cup all-purpose flour
2 medium onions, chopped
2 celery ribs, chopped
1 large green bell pepper, chopped
1/2 lb fresh or frozen pork tasso* (Cajun-cured smoked meat), thawed if frozen, trimmed and cut into 1/4-inch pieces
1 1/2 teaspoons salt
1/2 teaspoon cayenne
10 oz fresh or frozen baby okra, thawed if frozen, trimmed, and cut into 1/4-inch-thick rounds (2 cups)
3/4 cup thinly sliced scallion greens
Accompaniments: cooked white rice; hot pepper sauce

Simmer shrimp shells and water, uncovered, in an 8-quart pot until
liquid is reduced to about 12 cups, 15 to 20 minutes, then pour
through a sieve set over a large bowl and discard shells.
:
```

Figure 9: Contents of GUMBO1.TXT (inode 4) (E05).



```
SHRIMP AND ANDOUILLE SAUSAGE GUMBO

1/2 cup vegetable oil
1/2 cup all purpose flour
4 celery stalks, coarsely chopped
2 medium onions, coarsely chopped
2 green bell peppers, chopped
2 bay leaves
2 teaspoons salt
2 teaspoons dried oregano, crumbled
1/2 teaspoon cayenne pepper
5 8-ounce bottles clam juice
1 28-ounce can plum tomatoes, drained, chopped
1 pound smoked andouille or kielbasa sausage, halved lengthwise,
 sliced 1/4 inch thick
1/2 pound okra, trimmed, cut crosswise into 1/4-inch-thick slices

2 pounds uncooked medium shrimp, peeled, deveined
Freshly cooked long-grain rice
2 tomatoes, seeded, diced

Heat oil heavy large Dutch oven over high heat until almost
smoking. Add flour and stir until dark red-brown, about 8
minutes. Immediately add celery, onions and bell peppers. Cook 5
minutes, stirring and scraping bottom of pan often. Mix in bay leaves,
salt, oregano and cayenne. Add clam juice, canned tomatoes and
sausage. Boil 15 minutes. Add okra, reduce heat and simmer until okra
is tender, about 15 minutes. (Can be made 1 day ahead. Cover and
chill. Bring to simmer before continuing.)

Add shrimp to gumbo and simmer until just cooked through, about 3
minutes. Mound rice in soup bowls. Ladle gumbo over. Sprinkle with
```

Figure 10: Contents of GUMBO2.TXT (inode 6) (E05).

Interpretation. The volume was formatted with a tool that writes the OEM name “mkdosfs”. The visible content is unrelated to the case, so any images on the device exist only in unallocated space and were found by carving (Q3).

Evidence. E03, E04, E05.

Limit. FAT stores no time zone. Sleuth Kit labels the times “WAT” from the analysis VM’s zone, which says nothing about the device’s own zone, so they cannot be converted to UTC. The access date has day resolution only, and all FAT times can be altered. The unallocated area was examined only through carving.

Q3. What was recovered from the image?

Finding. PhotoRec 7.2 ran on the working copy (opened read-only, partition FAT16, 506,848 sectors) and reported 132 files saved in the first output folder (Figure 11). A second output folder, recup_dir.2, also exists (Figure 12).

The first folder contains seven JPEG files, two GIF files, one Word document and many .txt files (Figure 12). PhotoRec restores neither original names nor file-system timestamps, so names such as f0104057.jpg are offsets, not evidence of original naming.

Carved file	Type	Bytes	SHA-256 (first 12 hex)	Unique or duplicate	Notes
f0104057.jpg	JPEG	95,814	2261020a0b31	Unique	528 × 792; comment “copyright 2000 philg@mit.edu”
f0104249.jpg	JPEG	415,534	8cdb89c6778d	Unique	
f0105065.jpg	JPEG	411,361	452075313c93	Unique	

Carved file	Type	Bytes	SHA-256 (first 12 hex)	Unique or duplicate	Notes
f0105873.jpg	JPEG	264,600	f92654d9ee17	Duplicate of f0335081.jpg	Byte-identical pair
f0335081.jpg	JPEG	264,600	f92654d9ee17	Duplicate of f0105873.jpg	1024 × 768
f0106393.jpg	JPEG	6,809	568457ed594c	Unique	
f0106409.jpg	JPEG	230,665	b4f6bbb8d846	Unique	Same size as the archive entry rhino2.jpg (Q6)
f0106865.gif	GIF	11,407	8a67d4d406ed	Unique	
f0106889.gif	GIF	4,105	71e9f6b94e04	Unique	
f0335017_She_died_in_February_at_the_age_of_74.doc	Word 8.0	30,720	95c06c8815cf	Unique	See below

Table 3. Carved images and document (first 12 hex digits of each SHA-256; full values are in the hash listing, Figure 13).

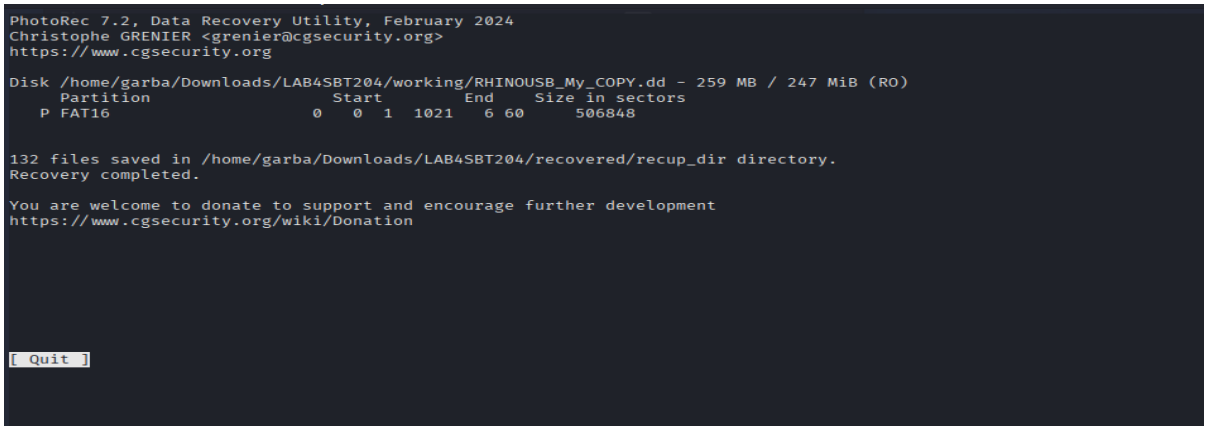


Figure 11: PhotoRec 7.2 summary on the working copy: FAT16, 506,848 sectors, 132 files saved (E06).

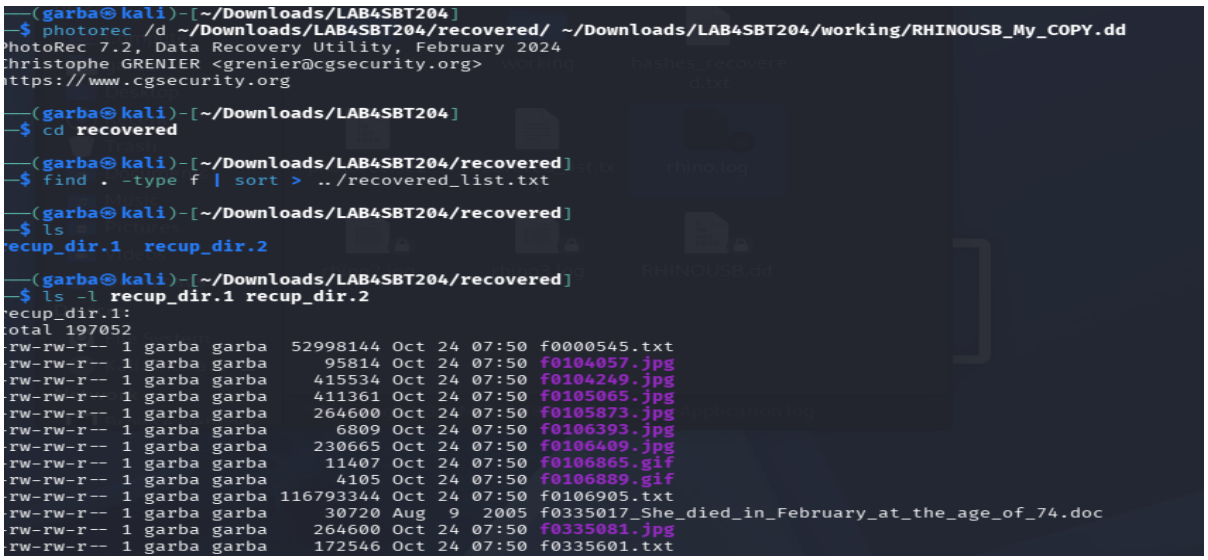


Figure 12: PhotoRec command, output folders recup_dir.1 and recup_dir.2, and listing of recup_dir.1 (E06).

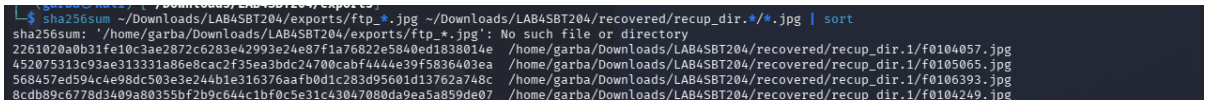


Figure 13: SHA-256 of the seven carved JPEG files (f0105873.jpg and f0335081.jpg identical); the ftp_*.jpg error is expected because no FTP objects had been exported yet (E07).

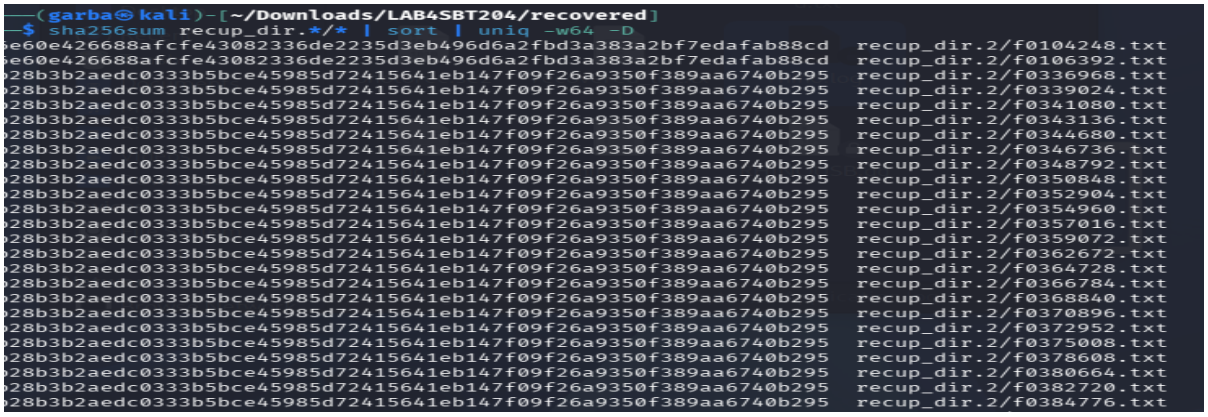


Figure 14: Duplicate-hash listing (uniq -w64 -D) of carved files: many identical .txt files in recup_dir.2 (E07).

Duplicates. f0105873.jpg and f0335081.jpg are byte-identical (264,600 bytes), so the JPEG set has six unique members (Figure 13). Carving cannot say whether the pair was two files on the device or one file stored twice. Many carved .txt files in recup_dir.2 share one hash (Figure 14). Two very large carved .txt files exist in recup_dir.1 (f0000545.txt, 52,998,144 bytes, and f0106905.txt, 116,793,344 bytes). These text files were not analysed as case evidence.

Word document. ExifTool reports title “She died in February at the age of 74”, author “NO WAY MAN NO WAY MAN NOWAY.”, company University of New Orleans, created 2005-08-09 02:17 and modified 02:40 (Figure 15). Its internal dates are later than every 2004 event in the captures, and nothing in the metadata ties it to the case, so it is recorded as an unlinked recovered artefact.

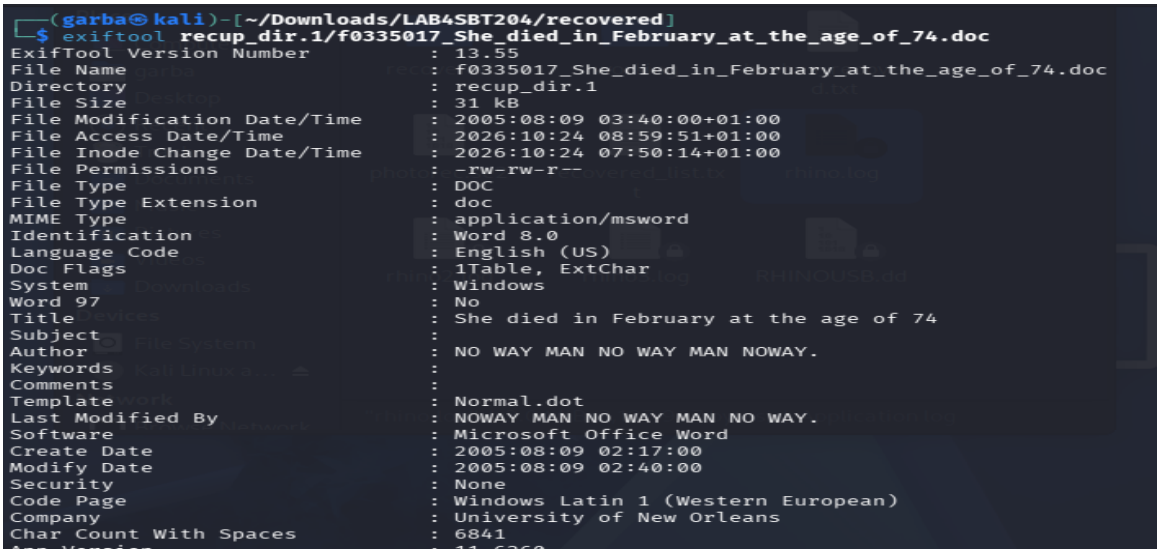


Figure 15: ExifTool 13.55 metadata of the carved Word document (E08).

Evidence. E06, E07, E08.

Limit. Carving recovers content, not names, paths or timestamps. File dates shown by `ls` for carved files come from PhotoRec's run time or from dates inside the file, so they are not creation evidence.

Q4. Is there hidden content in the recovered files?

Finding. None was found with the checks that could be run. A steganography detector could not be built, so the question is open for hiding methods that alter the image data itself.

Check	Tool and target	Result	Source
JPEG metadata	`exiftool *.jpg` (7 files read)	JFIF 1.01, baseline DCT with Huffman coding, YCbCr; no camera, software, GPS or date fields in the portions captured. f0104057.jpg is 528 × 792 with the comment "copyright 2000 philg@mit.edu".	Figure 16
Appended or embedded data	`binwalk f0104057.jpg`	One signature only: JPEG image data (JFIF 1.01) at offset 0. binwalk was run on this file only.	Figure 17
Readable strings	`strings -n 8 f0104057.jpg`	Only the comment string and compressed-data noise.	Figure 18
Steganography detection	stegdetect, built from source	Not performed. The build failed at configure, at the bundled jpeg-6b library and in the compile step; no binary was produced.	Figure 19 to Figure 22
Recovery of hidden content	none	Not attempted, because there was no detector result to justify it.	n/a

Table 4. Hidden-content checks and results.

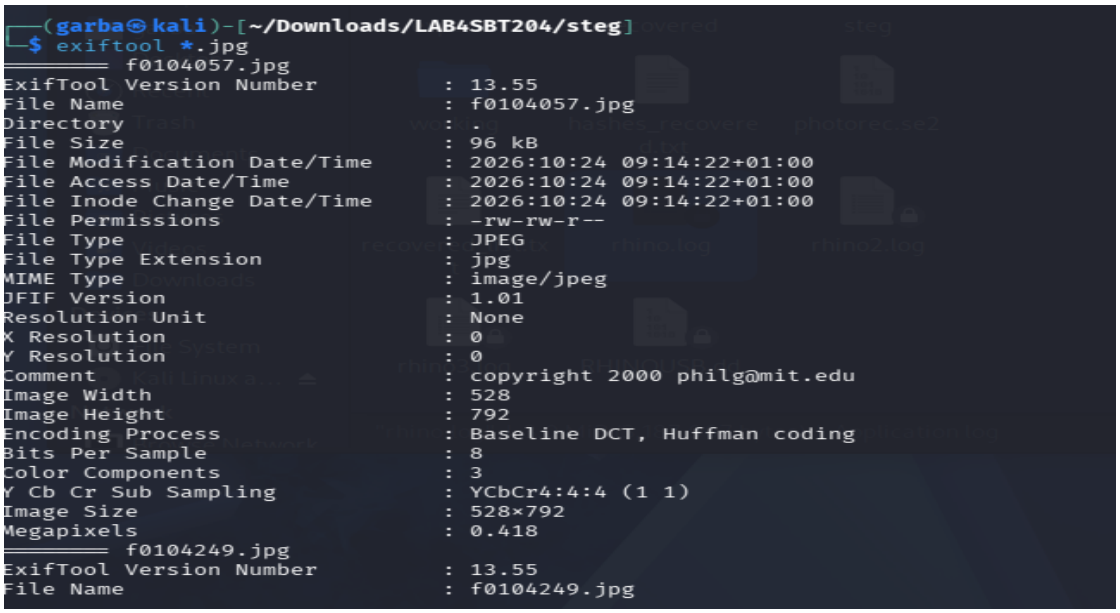


Figure 16: ExifTool output for the carved JPEG files (first file f0104057.jpg shown) (E09).

```
┌──(garba㉿kali)-[~/Downloads/LAB4SBT204/steg]
└─$ binwalk f0104057.jpg

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
0             0x0             JPEG image data, JFIF standard 1.01


┌──(garba㉿kali)-[~/Downloads/LAB4SBT204/steg]
└─$ strings -n 8 f0104057.jpg | less

zsh: done       strings -n 8 f0104057.jpg |
zsh: suspended  less
```

Figure 17: binwalk on f0104057.jpg: a single JPEG signature at offset 0 (E10).

```
copyright 2000 philg@mit.edu
 $.' ",#
(7),01444
'9=82<.342
!2222222222222222222222222222222222222222222222222
UE!vI-N=
:ChbP2+r
5LhI3Gnp
FRpO4S@h
v'hBivc^
qOi"l=Txdg
qRwtQuc;]8[
YKc$;cGH
h44YL6lPu
8=+Jl   "V
wbIb}MQ-
zZ9Wd@XFgeI
[#;oCc::
|gXemJf#j
G"K     +4Gp
{WBV3e:^
tQ{rfhcf6
^nGRg6I(
9-#uHss*
wERjFx|(
(4R3L^#(
kH<Xr|?x
2C+yYdEP
P[8POJ_c
Egf|I%By
\QG"E/8RA4
<n,|sqfu
:
```

Figure 18: strings -n 8 output for f0104057.jpg (E10).

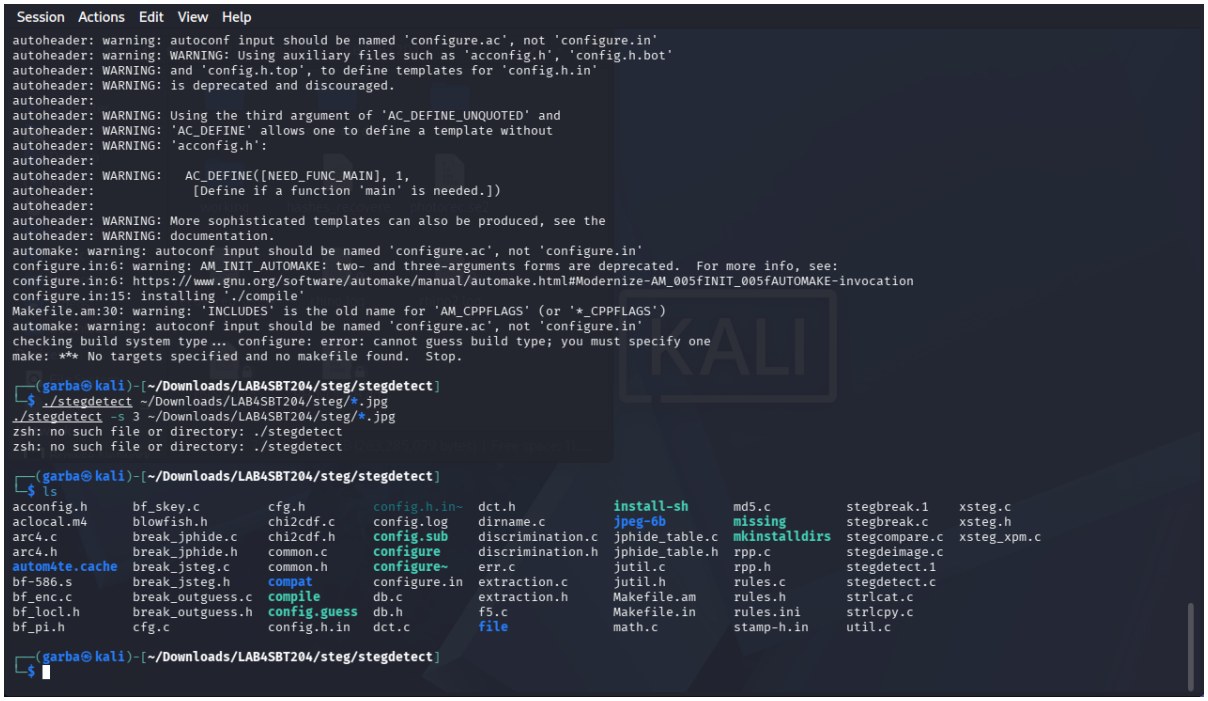


Figure 19: stegdetect build attempt 1: configure cannot guess the build type (E11).

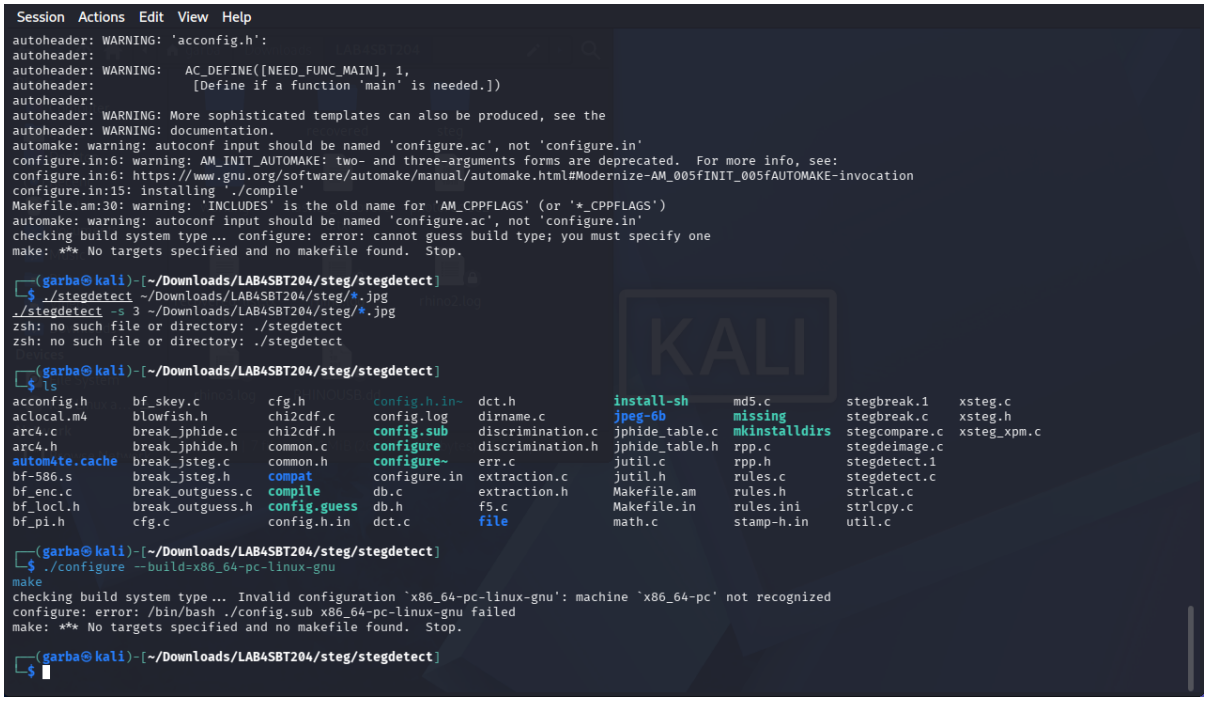


Figure 20: stegdetect build attempt 2: machine x86_64-pc not recognised by the bundled config.sub (E11).

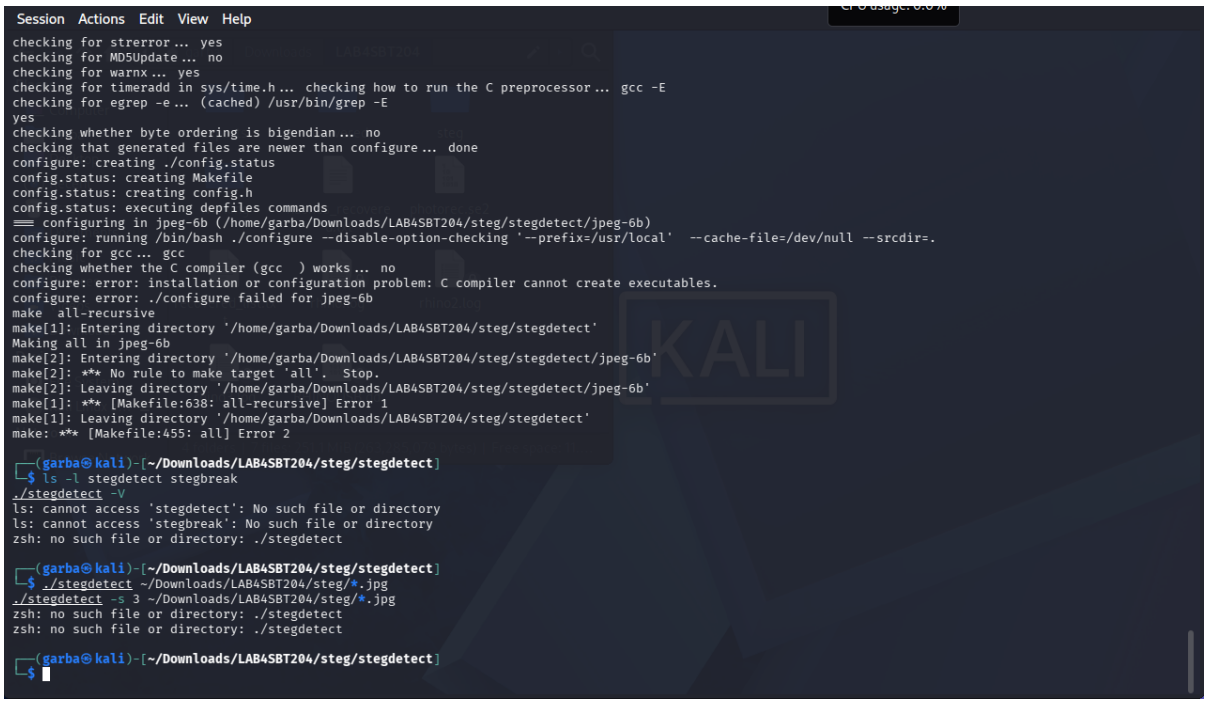


Figure 21: stegdetect build attempt 3: the bundled jpeg-6b configure fails (C compiler test) (E11).



Figure 22: stegdetect build attempt 4: stegdetect.c fails to compile; no stegdetect binary produced (E11).

What weakens a hiding interpretation: no appended signature, no unusual metadata and no readable payload in the file tested. **What supports one:** nothing observed. **Net:** inconclusive. Embedding inside the compressed image data would not appear in any of these checks, and the teaching deck's description of a hiding workflow is not treated as a finding of this report.

Evidence. E09, E10, E11.

Limit. Only f0104057.jpg was tested with binwalk and strings in the captured output, and metadata for the remaining files was reviewed in part. Absence of a result is not proof that nothing is hidden.

Q5. What does the FTP capture show?

Finding. rhino.log holds three FTP control sessions (TCP streams 69, 72 and 305) between the client 137.30.122.253 and the server 137.30.120.40 (banner "cook FTP server"). The account gnome logged in each time and uploaded files with STOR. All times are UTC on 2004-04-26.

Session	TCP streams	STOR and data port	Server replies and packets	Object, size and SHA-256
1	Control 69; data 71	Packet 1546, 22:21:49.954. PORT …6,121 = port 1657	150 BINARY at 1550; data 1551 to 1605; 226 at 1612 (22:21:50.179)	rhino1.jpg, 65,703 bytes, 0f30a8777d19a4fe…
2a	Control 72; data 79	Packet 1649, 22:22:08.905. PORT …6,124 = port 1660	150 ASCII at 1653; data 1654 to 1732; 226 with warning at 1737 (22:22:09.039)	rhino3.jpg (ASCII attempt), 96,899 bytes, a85aea8e47d9f542…
2b	Control 72; data 83	Packet 1763, 22:22:16.091. PORT …6,126 = port 1662	150 BINARY at 1767; data 1768 to 1846; 226 at 1851 (22:22:16.262)	rhino3.jpg (binary), 96,899 bytes, same SHA-256 as 2a
3	Control 305; data 307	Packet 5647, 22:26:46.671. PORT …7,210 = port 2002	150 BINARY at 5651; data 5652 to 5832; 226 at 5839 (22:26:46.876)	contraband.zip, 230,566 bytes, b7091a0b4295a561…

Table 5. FTP sessions, uploads and reconstructed objects (full hashes in Appendix B).

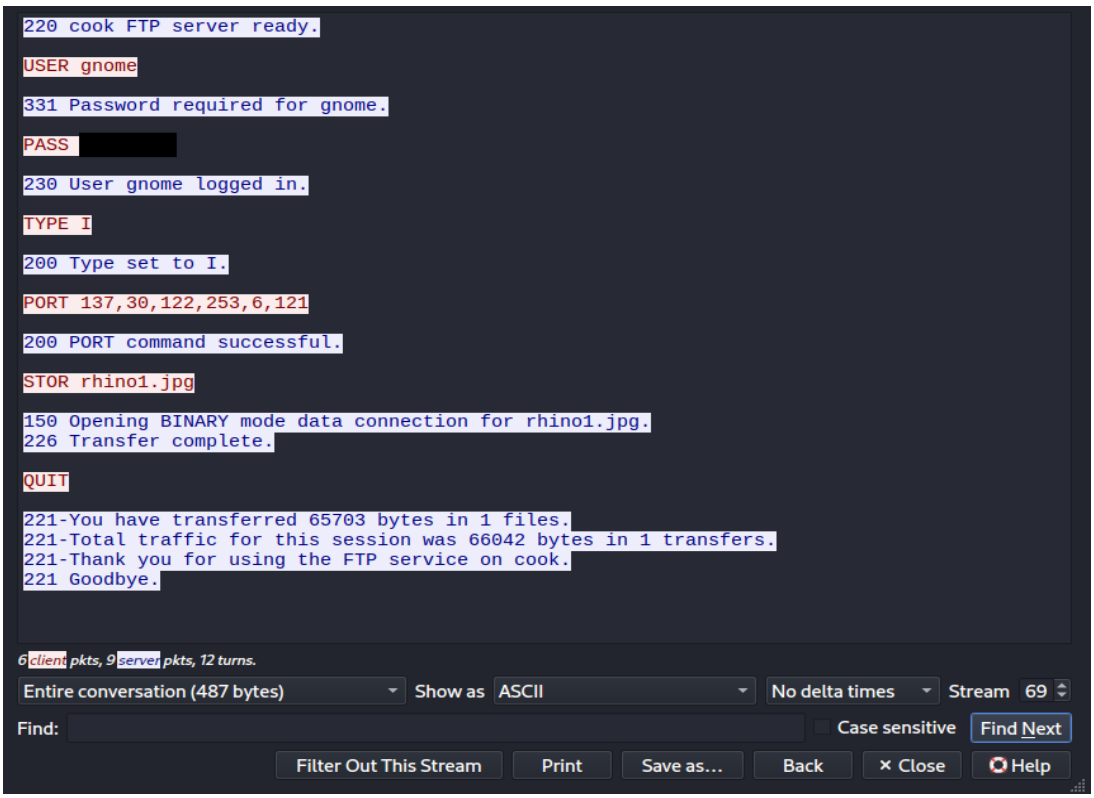


Figure 23: rhino.log, TCP stream 69 (FTP control, session 1): STOR rhino1.jpg, 150 BINARY, 226, 65,703 bytes transferred (password redacted) (E12).

- **Direction.** Every transfer is an upload (client to server). The server's byte counts match the reassembled data: 65,703 bytes in session 1 (Figure 23) and 230,566 bytes in session 3. Session 2 reports 193,797 bytes for two files, one byte less than twice the reassembled size (193,798); this was not resolved.
- **ASCII-mode attempt.** The first rhino3.jpg transfer used ASCII mode and the server replied that 321 bare linefeeds were received and the file may not have transferred correctly. The client then switched to binary (`TYPE I`) and sent the file again. The two captured data streams are byte-identical and both are valid JPEG data (`FFD8` to `FFD9`), so the capture does not show corruption of what the client sent. The warning concerns the server's stored copy, which is not in evidence.
- **Date.** The first transfer began at 22:21:49.954 UTC on 2004-04-26, as shown by the capture. This differs from the date quoted in the teaching deck (2004-04-20), which was not used.
- **Credentials.** FTP is unencrypted, so the user name and password are visible in the capture. The password is redacted in this report and was not reused.

Limit. The capture shows bytes sent from endpoint 137.30.122.253 under the account gnome and the server's completion messages. It does not identify the person, and it does not show what the server stored.

Q6. What is the protected archive and what does it hold?

Finding. The reassembled contraband.zip is 230,566 bytes, begins with the ZIP signature `PK 03 04`, contains an end-of-central-directory record and matches the server's byte count, so it was captured completely. It has a single entry, rhino2.jpg (230,665 bytes uncompressed, 230,448 compressed, deflate), and the encryption flag is set on that entry. The entry's compression method is deflate (not method 99), which is consistent with the traditional ZIP password scheme and not AES.

- **First save attempt.** The first file saved as ftp_contraband.zip was the FTP control stream (`file`: "ASCII text", beginning "220 cook FTP server ready"), and `fcrackzip` rejected it as not a zipfile (Figure 24, Figure 25). The archive was then saved from the data connection (port 2002, stream 307).

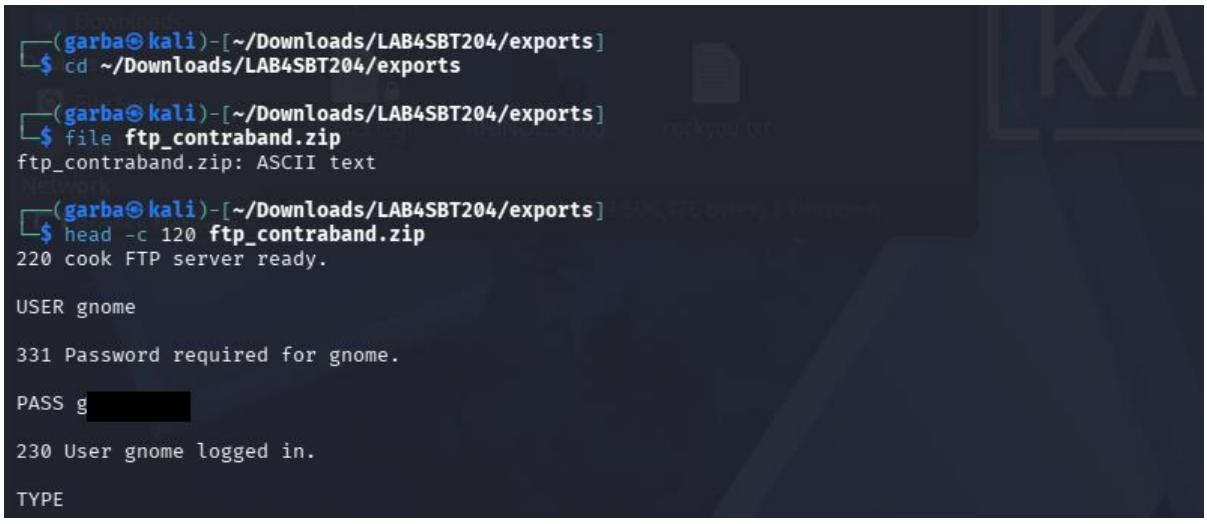


Figure 24: First attempt to save contraband.zip captured the FTP control stream (ASCII text), not the archive (password redacted) (E15).

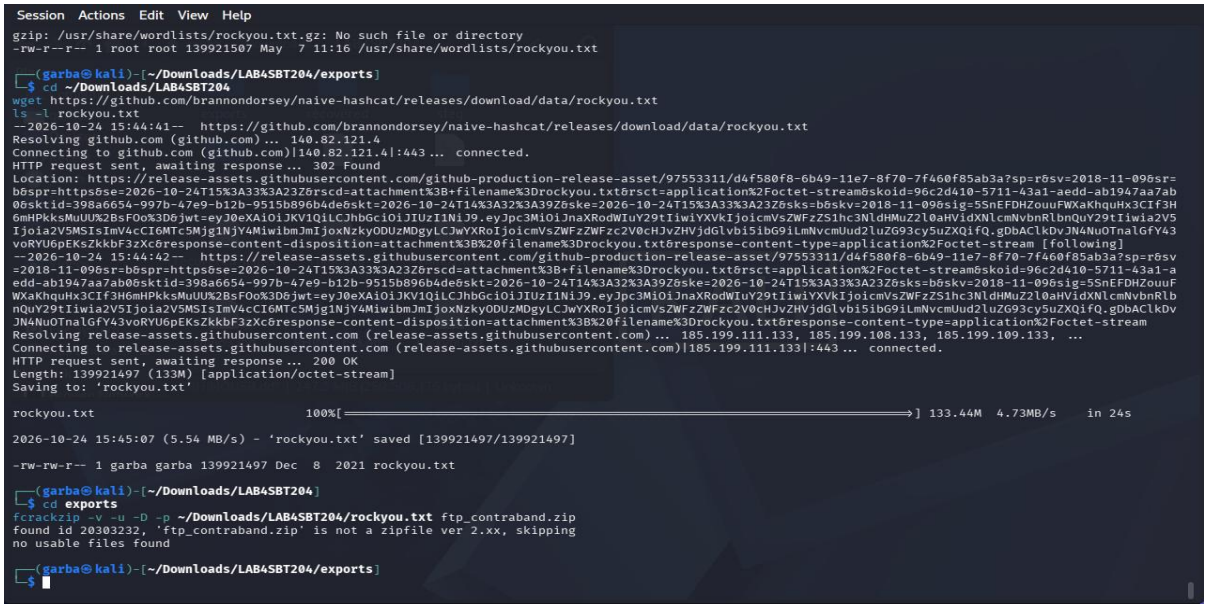


Figure 25: fcrackzip rejects the mis-saved file (not a zipfile); wordlist rockyou.txt present (E15).

Limit. Opening the archive shows that the password was recovered, not who created it or who knew the password.

Q7. What does the HTTP traffic show?

Finding. In both captures the client is 137.30.123.234 and the web server is 137.30.120.37 (`Host: www.cs.uno.edu`). All times are UTC on 2004-04-28.

Request / response packet	UTC time of request	Endpoints and stream	Host and URI	Response	Object check
49 / 50 (rhino2.log)	21:08:38.073	137.30.123.234 → 137.30.120.37:80, stream 1	www.cs.uno.edu /~gnome/rhino4.jpg	200 OK, image/jpeg, 153,191 bytes	FFD8 … FFD9; e1c01da39166…
217 / 219 (rhino2.log)	21:08:44.189	same endpoints, stream 2	www.cs.uno.edu /~gnome/rhino5.gif	200 OK, image/gif, 85,137 bytes	GIF89a … 3B; f36ecc12967d…
110 / 112 (rhino3.log)	21:16:23.343	same endpoints, stream 3	www.cs.uno.edu /~gnome/rhino.exe	200 OK, application/octet-stream, 145,920 bytes	MZ (PE32); 93e70049b60b…

Table 6. Material HTTP requests and responses. For each, the exported body length equals the Content-Length.

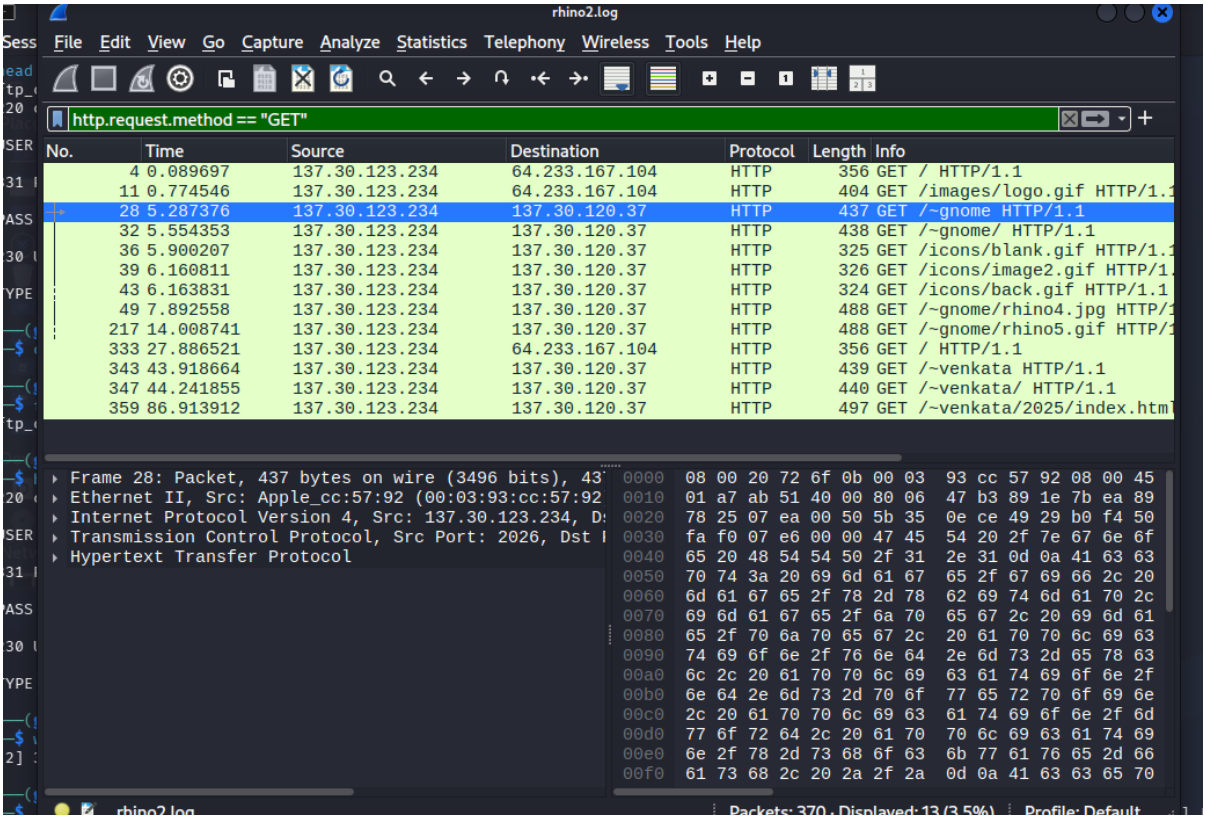


Figure 26: rhino2.log filtered with http.request.method == "GET": 13 of 370 packets (Time column shown as seconds since capture start) (E17).

- **Excluded requests.** The other requests in rhino2.log (Google home page and logo, /~gnome directory pages, the icons blank.gif, image2.gif and back.gif, and /~venkata pages) are page furniture or browsing and were reviewed and excluded as case objects.
- **Sequence in rhino3.log.** Before the download, the client ran a Google web search for “rhino.exe” (21:15:48.234) and a Google Groups search for the same term (21:15:55). It then requested `/~gnome/rhino.exe` at 21:16:23.343 (Figure 28).
- **Executable.** The response is a complete Windows PE32 executable (`MZ` header, 145,920 bytes). It was hashed and described only. Its behaviour is unknown because it was never run, so the capture proves a download, not execution.
- **Email traffic.**
- **Failed first export.** The first export attempt produced files named http_rhino4.jpg.pcap and http_rhino5.gif.pcap, which are not images and could not be decoded (Figure 27). They were discarded and the objects were re-exported (E20).



Figure 27: First export attempt: files named *.pcap are not images and fail to decode; they were not used as evidence (E18).

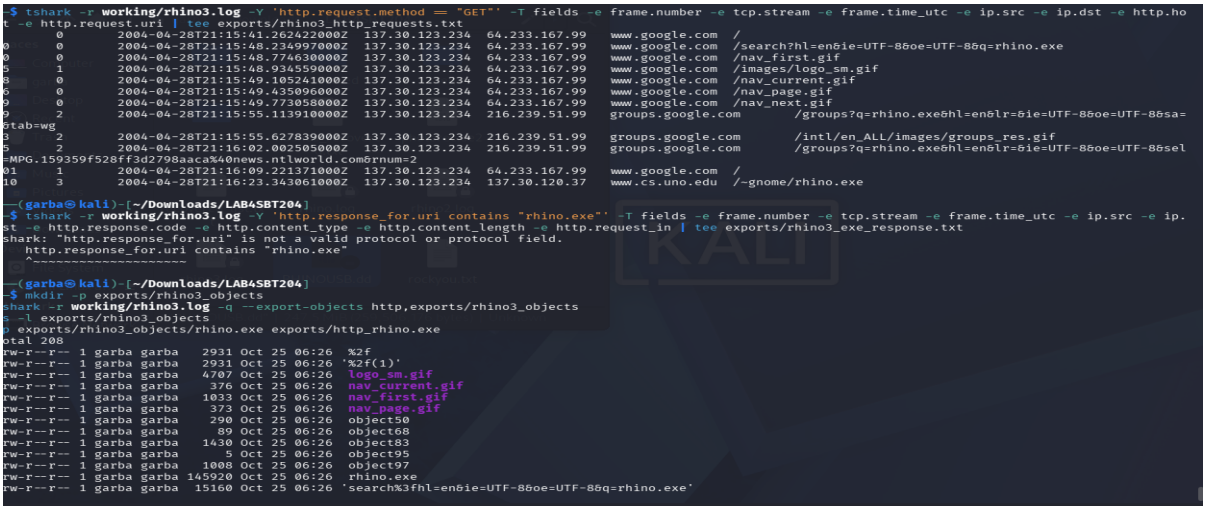


Figure 28: rhino3.log: tshark request list with UTC times, and HTTP object export (rhino.exe, 145,920 bytes) (E19).

Limit. The HTTP client (137.30.123.234) is a different endpoint from the FTP client (137.30.122.253), and the web server (137.30.120.37) differs from the FTP server (137.30.120.40). A registry lookup of an address would identify an address holder only, not the person who used it.

Q8. What can defensibly be concluded?

Conclusion. The evidence supports a narrow account: an endpoint using the account gnome uploaded three objects by FTP on 2004-04-26, a second endpoint downloaded two images and an executable from a university web server on 2004-04-28, and a USB image holds two recipe files plus carved remnants. It does not show that any carved file was transferred, that the two endpoints are related, or who operated either one.

Directly observed	Inferred (with limits)
Three FTP sessions, four STOR commands, server completion messages and byte counts (packets and streams in Table 5). Three HTTP downloads with status 200 and matching lengths (Table 6). Two allocated recipe files on a FAT16 volume; carved JPEG, GIF and Word content. No hash match between any FTP or HTTP image and any carved image.	The gnome account at 137.30.122.253 sent the uploaded files; the capture shows bytes sent, not what the server kept. The client at 137.30.123.234 requested rhino.exe after searching for it; execution is unknown. The shared label “gnome” (FTP user and /~gnome path) is consistent with one user area, but the servers differ and no packet links them. Carved images are remnants of files once on the device; their origin and any link to the transfers are unknown.

Table 7. Observations versus inferences.

Object	Bytes	SHA-256 (first 12 hex)	Matches a carved image?
rhino1.jpg (FTP)	65,703	`0f30a8777d19`	No
rhino3.jpg (FTP, both transfers)	96,899	`a85aea8e47d9`	No
rhino4.jpg (HTTP)	153,191	`e1c01da39166`	No
rhino5.gif (HTTP)	85,137	`f36ecc12967d`	No
rhino2.jpg (entry in contraband.zip)	230,665		Size equals f0106409.jpg, a lead only

Table 8. Cross-match of reconstructed network images against the carved images (by SHA-256 prefix; the unique images are six carved JPEGs, two carved GIFs, rhino1.jpg, rhino3.jpg, rhino4.jpg and rhino5.gif).

Confidence. High for the recorded transfers and objects. Low for any connection between the USB image and the network activity. Not supported for any named person. Limitations are listed in section 7.

TIMELINE AND CORRELATION

All times are UTC unless a row says otherwise. Rows are ordered by time. Rows 1 to 8 come from the packet captures, row 9 from the FAT file system (device-local time, zone unknown) and rows 10 and 11 from the examination itself. Packet and stream numbers are Wireshark values and can be checked in the capture.

STAGE	TIME (UTC)	EVIDENCE	SOURCE ARTEFACT	OBSERVED FACT	INTERPRETATION AND LIMIT
FTP TRANSFER ACTIVITY	2004-04-26 22:21:49.954	E12, E13, E14	rhino.log, control stream 69, packets 1546 to 1612	STOR rhino1.jpg (packet 1546); 150 BINARY (1550); data stream 71 (1551 to 1605); 226 at 22:21:50.179 (1612); server reports 65,703 bytes.	Upload from 137.30.122.253 to 137.30.120.40 under account gnome. Shows bytes sent and the server's completion message; not who sent them or what the server kept.
FTP TRANSFER ACTIVITY	2004-04-26 22:22:08.905	E13, E14	rhino.log, control stream 72, packets 1649 to 1737	STOR rhino3.jpg in ASCII mode (1649); 150 ASCII (1653); data stream 79; server warning at 1737 (22:22:09.039): 321 bare linefeeds, file may not have transferred correctly.	Captured bytes are a valid 96,899-byte JPEG. The warning concerns the server's copy, which is not in evidence.
FTP TRANSFER ACTIVITY	2004-04-26 22:22:16.091	E13, E14	rhino.log, control stream 72, packets 1755 to 1851	TYPE I, then STOR rhino3.jpg in binary mode (1763); 150 BINARY (1767); data stream 83; 226 at 22:22:16.262 (1851).	Second transfer is byte-identical to the first (same SHA-256), so one unique image was sent twice.
FTP TRANSFER ACTIVITY	2004-04-26 22:26:46.671	E14, E16	rhino.log, control stream 305, packets 5647 to 5839	STOR contraband.zip (5647); 150 BINARY (5651); data stream 307 (5652 to 5832); 226 at 22:26:46.876 (5839); 230,566 bytes; one encrypted entry, rhino2.jpg.	Complete archive. Contents depend on the password recovery result (Q6).
HTTP TRANSFER ACTIVITY	2004-04-28 21:08:38.073	E17, E20	rhino2.log, stream 1, packets 49 and 50	GET /~gnome/rhino4.jpg from 137.30.123.234 to www.cs.uno.edu (137.30.120.37); 200 OK, image/jpeg, 153,191 bytes (response at 21:08:38.099).	Download of one JPEG by the HTTP client. Shows retrieval, not viewing or storage.
HTTP TRANSFER ACTIVITY	2004-04-28 21:08:44.189	E17, E20	rhino2.log, stream 2, packets 217 and 219	GET /~gnome/rhino5.gif; 200 OK, image/gif, 85,137 bytes (response at 21:08:44.221).	Second download six seconds after the first, same client and server.
HTTP TRANSFER ACTIVITY	2004-04-28 21:15:48.234	E19	rhino3.log, stream 0, packet 10	Google web search /search?q=rhino.exe from 137.30.123.234; a Google Groups search for the same term follows at 21:15:55.	The client searched for the executable before requesting it. Search intent is not proven by the search itself.
HTTP TRANSFER ACTIVITY	2004-04-28 21:16:23.343	E19, E20	rhino3.log, stream 3, packets 110 and 112	GET /~gnome/rhino.exe; 200 OK, application/octet-stream, 145,920 bytes (response at 21:16:23.384); MZ header, PE32.	Complete download of a Windows executable. Not run during this examination, so its function is unknown.
DISK IMAGE REVIEW	2004-04-30 18:11:20 and 18:11:24 (device-local, zone unknown)	E04	RHINOUSB.dd, inodes 4 and 6	GUMBO1.TXT written and created 18:11:20; GUMBO2.TXT written and created 18:11:24.	FAT stores no time zone, so these cannot be converted to UTC. If the device clock was comparable they are later than every network event; they relate to recipe files, not to images.
MEDIA PRESERVATION	2026-10-23 22:05 (23:05 local, +01:00)	E01	Originals in ~/Downloads/LAB4SBT204/	Last-modified time listed for the four originals when they were made read-only; SHA-256 values recorded before analysis.	This is the time of the listing, not of the original acquisition. The offset is the one ExifTool reports for the analysis VM.

RECOVERED CONTENT	2026-10-24 06:50 (07:50 local, +01:00)	E06, E08	recovered/recup_dir.1	PhotoRec output files written (listing time 07:50; inode change time 07:50:14 +01:00).	Shows when carving ran, not when the carved files were created.
CORROBORATED CONCLUSION	n/a (analysis)	E01 to E21	All artefacts	Transfers and downloads are established by packets and hashes. No FTP or HTTP image matches a carved image by SHA-256.	The USB image and the network activity are not linked by this evidence, and no person is identified.

Table 9. Chronological timeline (events from at least three artefact families: packet captures, file system and recovered content, and examination records).

6.1 Correlation across artefacts

- **Corroborates:** the FTP server's own byte counts (65,703 and 230,566) equal the reassembled streams, and each HTTP Content-Length equals the exported body length. These are independent checks of the same objects.
- **Consistent but not conclusive:** the label "gnome" appears as the FTP account and as the web path /~gnome. The FTP server (137.30.120.40) and the web server (137.30.120.37) differ, and no packet shows that uploaded files were the ones served.
- **Does not fit or does not establish:** the FTP client (137.30.122.253) and the HTTP client (137.30.123.234) are different endpoints; no carved image matches any transferred image by SHA-256; and the FAT times (2004-04-30) fall after all captured network events but belong to the two recipe files, so they say nothing about when the carved images were written.

CONCLUSION AND LIMITATIONS

The evidence establishes three FTP uploads and three HTTP downloads at the times, endpoints and sizes recorded above, and a FAT16 USB image whose only visible files are two recipes. It does not establish possession by a named person, completed transfer to a named person, or any intent. The confidence ratings are those in section 2.

Limitations

1. **Account and endpoint are not a person.** The captures show an account name (gnome) and IP addresses. Shared or unattended machines, shared credentials and address reuse cannot be excluded.
2. **Two different endpoints.** The FTP client (137.30.122.253) and the HTTP client (137.30.123.234) differ. Nothing in the evidence shows they are the same user or machine.
3. **No link to the USB image.** The carved images do not match any transferred image by hash, and the visible files are recipes. Carved content has no names, paths or timestamps, and carving can produce fragments.
4. **Server-side state unknown.** The captures show bytes sent and server messages, not what the server stored. The ASCII-mode warning relates to the server's copy.
5. **Archive.** The encrypted entry depends on a password recovery performed on a training archive in the course lab. Success shows the password was found, not who set or used it.
6. **Timestamp uncertainty.** FAT times have no time zone and can be altered; packet times depend on the capture host's clock; the teaching deck's date for the first transfer (2004-04-20) conflicts with the capture (2004-04-26) and was not used.
7. **Hidden-content testing incomplete.** The steganography detector would not build, binwalk and strings were run on one file, and DCT-domain hiding would not appear in the checks that were run.
8. **Duplicates and noise.** One JPEG was carved twice and many carved text files are identical; counts of "files" therefore overstate unique evidence.
9. **Executable not run.** The download of rhino.exe is proven; its behaviour and any execution are unknown.
10. **Hash equality is not origin.** Matching hashes show identical bytes, not who created or first held a file.

Alternative explanations

The recovered remnants could be earlier contents of a reused device and unrelated to the 2004 network activity. The two clients could belong to different people sharing an account name. The searches and download in rhino3.log could reflect research or testing rather than possession. The

evidence cannot distinguish these from the account in section 2 without further sources such as server logs, the server's stored files or device-ownership records.

ACADEMIC INTEGRITY AND AUTHORIZATION STATEMENT

I confirm that I completed this digital forensics case study in an authorized virtual environment using the supplied scenario dataset. The commands, record citations, evidence-log entries, cryptographic hashes and observations in this report are my own work and are accurately documented, including the handling error and the failed tool build described above. Course teaching materials were used as guidance only, and their stated results were not reported as my findings.

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Date: October 25, 2026